DEPT. OF CHEMICAL ENGINEERING

CH 5440 - MULTIVARIATE DATA ANALYSIS

END SEMESTER EXAM: TIME: 9AM-1PM DATE: 06/05/23

INSTRUCTIONS

1. Open book and notes exam. You are free to consult your notes, text books, research papers, internet for solving the problems. You can reuse the codes you have developed for solving the assignments.
2. You are encouraged to use MATLAB or Python to solve the problems. Attach a print-out of the code (should be well documented) along with your submission.
3. **The answers should be provided in a separate word or pdf document. The document along with the MATLAB or PYTHON codes should be uploaded as a single zip file. Name the zip file as rollnumber.zip. Name each MATLAB/PYTHON file as Qxx.m or Qxx.ipynb.**

Before you begin to answer the questions

For the three problems MATLAB codes are given which should be used to generate individualized data sets. These MATLAB codes are respectively *generate_network_data.m*, *generate_reaction_data.m* and *generateDAE_RC_data.m*. The data files that will be generated by these codes are respectively *networkdata.mat*, *reactiondata.mat* and *DAEdata.mat*. **Before executing each of these codes change the random number used in line 4 to your roll number. For example if your Roll number is CH20B018 then the random number to be used is 20018, that is, line 4 should be changed to *rng(20018,'twister')*.**

1. The file networkdata.mat contains 1000 samples of flow measurements of some flow network. The variable Ftruer contains the true values and the variable Fmeasr contains the measured values. Assume the errors in all flow measurements are independent and have same variance.

(a) Apply PCA to the **true flow data** to determine the true number of constraints and the true constraint matrix relating the flow variables. Report the number of constraints and provide a reason for your answer. (5 points)

(b) Use the true constraint matrix obtained in part (a) to choose a valid set of dependent variables and report them, along with an explanation for your choice. Determine the true regression matrix relating the dependent flows to the independent flow variables for your choice. (10 points)

(c) Apply PCA to the **measured flow data** and estimate the number of constraints as well as the constraint matrix. Justify your selection of the number of constraints based on hypothesis testing for equality of smallest eigenvalues. Report the test statistics and test criterion for different guesses of number of constraints between 12 and 5 in the form of a table. Report the estimated number of constraints and estimated error variance. (10 points)

(d) Assume that the true number of constraints is known (as obtained from part (a)). For the same choice of dependent variables as chosen in (b) estimate the regression matrix relating the dependent to the independent flow variables using the estimated constraint matrix in part (c). Report the maximum absolute difference between the estimated and true regression matrix elements. (5 points)

2. Consider the problem of developing the kinetic rate equation from experimental data. The kinetic rate equation relates the time rate of change of concentration (C_A) of a species A as a function of temperature (T) and concentration of the species. It is given as follows

$$r_A = -\frac{dC_A}{dt} = kC_A^n$$

The rate constant *k* is a function of temperature and is given by

$$k = k_0 e^{-\frac{E}{T}}$$

The parameters k_0, E and n are unknown and have to be estimated from data. Measurements for the concentration of species A with respect to time have been obtained from experiments conducted at several different temperatures and are stored in file *reaction.mat* (variable t stores the time instants in min when concentrations are measured, variable Ca is a matrix of concentration measurements (in gmol/l) where each column corresponding to concentrations at different time instants for a fixed temperature, and temp contains the different temperatures in deg K at which experimental data are obtained). Assume concentrations measurements are noisy and time and temperature measurements are non-noisy.

(a) Use OLS to obtain the rate constant k and order n from each experimental data conducted at a fixed temperature. For this purpose, approximate the derivative at each time instant using forward difference. From the rate constants estimated at different temperatures obtain an estimate of k_0 and E using OLS. Report the estimated values of rate constants, mean value of the reaction orders n, k_0 and E (10 points)

(b) Using the estimated values of *k* obtained in (a), apply OLS to obtain an unique estimate of *n* using all the concentration measurements (10 points)

(c) Assuming reaction order n is greater than 1, integrate the differential equation at a fixed temperature to obtain a nonlinear relation for concentrations as a function of time. Use nonlinear OLS to estimate the parameters k and n from the data for each fixed temperature From the rate constants estimated at different temperatures obtain an estimate of k_0 and E using OLS. Report the estimated values of rate constants, mean value of reaction orders n, k_0 and E (10 points)

(d) For the above models, report the concentration of the species at t = 15 mins, if the reaction is conducted at a temperature of 500 deg K starting with an initial concentration of 1.0 gmol/lit. For parts (a) and (c) use the average of the estimated values of n to obtain a single model (5 points)

Note: For solving parts (a) and (b) you need to convert the problem to a linear regression problem using a suitable transformation, if necessary.

3. A linear descriptor system is one which can be described by a differential algebraic model, that is, a model consisting of a number of linear difference equations and a number of linear algebraic equations relating the input and output variables. One simple example is that of a resistor-capacitor circuit typically used to develop battery models, which can be modelled **using one difference equation and several algebraic equations**. 1024 measured data for an RC circuit is given in DAEdata.mat. The variable ymeas contains the measured values of output variables and variable umeas contains the measured values of the input variable. The standard deviations of errors in the output and input measurements are respectively given in the variables stdey and stdeu, respectively.

(a) Assuming that the error standard deviations are known, determine the number of algebraic equations relating the variables. Describe the method you are using and report the number of algebraic constraints and corresponding constraint matrix. (5 points)

(b) Assuming that the error standard deviations are known, determine the order of the difference equation and the difference equation parameters. Describe the procedure you are using and report the difference equation parameters. (15 points)

(c) Assume that the number of algebraic constraints and error standard deviations are unknown, estimate the error variances and number of algebraic constraints. Describe the method you are using and report the estimated number of algebraic constraints and error standard deviations. (15 points)

Note: The relation between lag, order and number of constraints developed for a dynamical system have to be suitably modified for a linear descriptor system. Wherever required use hypothesis test to estimate the number of constraints.